

AAKASH PATHAK

Software Engineer | Full Stack & AI-Powered Systems

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with hands-on experience building production web systems and AI-powered platforms using Python, JavaScript, React, and Node.js. Experienced designing and consuming REST APIs, debugging across the stack, and building OCR/NLP pipelines with structured data extraction. Comfortable working end-to-end on scoped problems — from architecture through implementation, testing, and iteration — with a strong foundation in data structures, algorithms, and system design.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), Java, C++, HTML5, CSS3

AI / NLP: OCR/NLP Pipelines, Structured Data Extraction, JSON Schema Validation, Prompt-Based Processing

Backend: Node.js, Express.js, REST API Design, JWT Authentication, WebSockets (Socket.io)

Frontend: React.js, Tailwind CSS, Vite, Responsive Design, Component Architecture

Database: MongoDB, Mongoose, Schema Design

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

Tools & Platforms: Git, GitHub, Vercel, Render, Postman, VS Code

EXPERIENCE

Full Stack Developer Intern — Macziom Architects & Constructions

Apr 2026 – Jul 2026

Pitampura, Delhi

- Diagnosed page-load latency on the Build Buddy platform and shipped component-level code splitting and API-layer optimizations, improving load performance by ~30% and data-fetch latency by ~25%.
- Designed and integrated 10+ RESTful API endpoints connecting frontend interfaces with backend services, debugging issues end-to-end across the stack.
- Built a modular, reusable React component architecture across 5+ feature modules, cutting development time by ~40% for subsequent features.

UI Designer — Gettobyte Technologies Pvt. Ltd.

Nov 2024 – Dec 2024

Noida, India

- Collaborated with frontend engineers to deliver pixel-perfect implementation of design systems, reducing design-to-dev handoff iterations by ~50%.

UI/UX Designer — Banao Technologies

May 2024 – Jul 2024

Remote

- Partnered with engineering teams to ship UI solutions across 3 client products on web and mobile, improving task-completion flows through structured feedback cycles.

PROJECTS

DenkenAI — Python, Node.js, MongoDB, OCR, NLP | denken-ai.vercel.app | AI-Powered Academic Platform

- Built a Python-based OCR/NLP pipeline processing 100+ page PDFs to auto-generate structured exam questions with ~90% context accuracy, enforcing strict JSON schema validation on all pipeline outputs.
- Developed a performance analytics module evaluating test attempts across adaptive difficulty levels, reducing average study planning time by ~35% through personalized recommendations.

Eidos — React, Node.js, Playwright | bwp-eidos.vercel.app | Design Extraction Engine

- Built a Playwright-based headless automation layer performing real-time DOM traversal and CSS parsing to extract typography, color systems, and layout hierarchy from any live site in under 3 seconds.
- Debugged and validated rendering compatibility across 95%+ of modern web architectures, including JavaScript-heavy SPAs; deployed on Vercel.

KOTObA — React, Node.js, Socket.io | bwp-kotoba.vercel.app | Real-Time Multiplayer System

- Engineered a stateful real-time multiplayer game engine supporting 50+ concurrent players via WebSocket-based state synchronization, with live scoring and leaderboard tracking.
- Built a scalable game-state architecture managing timed rounds and concurrent sessions with <100ms event propagation latency; owned the system end-to-end from design to production deployment.

EDUCATION

B.Tech, Electronics and Communication Engineering — JSS Academy of Technical Education

2023 – 2027

Noida, India

CERTIFICATIONS & ACHIEVEMENTS

- Smart India Hackathon (SIH) Qualifier (2024) — Advanced to the penultimate round among 1M+ participants; led technical solution design and shipped a working prototype under strict time constraints.
- Full Stack Web Development — Udemy (2024)
- UI/UX Design — Udemy (2024)